

# Data Analysis Prelim Exam

Department of Statistics, University of Chicago

Thursday, Sept 19, 2019: Due Friday, Sept 20 at 4.00pm

## 1 Data

The data set for this exam is taken from the paper

*Effects of climate warming on photosynthesis in boreal tree species depend on soil moisture*, Reich et al, *Nature* vol. 562, pages 263–267 (2018)

The data are provided in the file `borealwarming.csv`.

**Experiment design** The paper studies the effect of warmer temperatures on photosynthesis in trees. Two sites a large distance apart were split into three blocks each, and each block was split into four plots. In each plot, warming cables were laid in the soil with the capability of maintaining a temperature that is 3.4 degrees Celsius warmer than the ambient temperature. Each plot was then randomly assigned to one of two treatments—**warmed**, where the underground cables and above-ground heating lamps were turned on during the growing season, or **ambient**, where the electrical cables were not turned on. Within each plot, trees from 11 species were planted and measurements were taken for the photosynthesis activity level (measured in a single leaf). The soil water level was also measured.

**Variables** The principal variables are:

- **site**, **block**, **plot\_id** denote the location of the plant being measured. There are two sites, six blocks (three per site), and 24 plots (four plots per block).
- **warming\_treatment** takes values **warmed** or **ambient**, for the treatment or control group. The randomization is at the level of the plot (i.e., all trees in the same plot are assigned the same treatment).
- **species** takes 11 values. Four species (**abiba**, **picgl**, **pinba**, **pinst**) are needle leaved (coniferous) trees, while seven species (**aceru**, **acesa**, **betpa**, **poptr**, **quema**, **queru**, **rhaca**) are broadleaf (deciduous) trees.
- **year** is the year when the measurement is taken (2009, 2010, or 2011).
- **doy** is the day of the year (e.g., **doy**=1 means that the measurement was taken on January 1st of that year). Note that two measurements may have the same **doy** value even if they occur in different years.
- **Asat** measures the photosynthesis activity of the leaf; 2049 measurements.
- **soil\_water\_VWC** measures the soil water content in the plot. (On any given day, this measurement is the same for all plants in the same plot.)
- You may ignore the other variables present in the data set.

## 2 Questions

You are encouraged to read the *Nature* paper before answering the questions, and you are not restricted to the specific questions mentioned below; you are free to examine other points of interest if you like. Throughout, you may set aside missing response values, and you may assume that measurements are independent across different days.

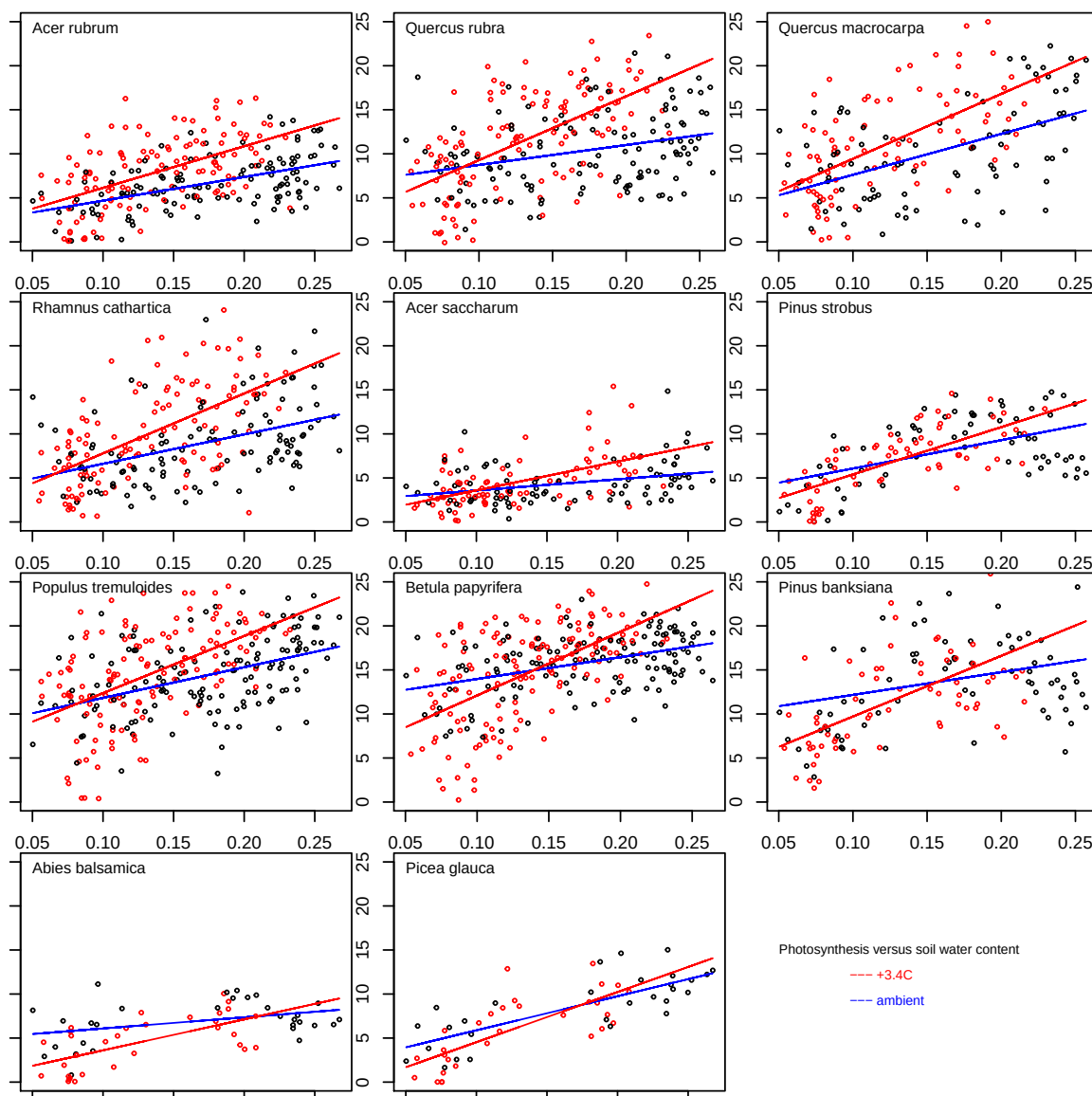
Your report should include code for all plots and results that you present.

1. Before any trees were planted, a grid of underground electrical cables was laid down at a depth of 15 cm in each plot, with sufficiently small separation that the heating effect could be deemed uniform. The cables in the treated plots were used as heating elements. What was the purpose of the cables in the control plots?

For comparative purposes, it is essential that the treated plots be as similar as possible to the control plots. Site preparation and tillage affects the soil—its friability, its drainage capacity, and possibly its solar exposure. If the control plots were not prepared and cabled in the same way as the treated plots, it could legitimately be argued that any observed difference in photosynthesis or leaf conductance is not due to treatment but to differences in site preparation. Ideally, plots must be prepared prior to randomization, which implies that cables are in place in all plots.

The purpose of cables in the control plots is to ensure comparability and to forestall objections about the lack thereof.

2. Use the data to reproduce the authors' plots in Fig. 1 in a similar format. What, if anything, do these plots tell you about the effect of elevated temperature on the various species? (Note that the species can be grouped into deciduous vs. coniferous trees as given above.)



The four coniferous species are the two pinus varieties plus abies and picea (pine, fir and spruce). The scatterplots for five of the seven broadleaf varieties look fairly similar in terms of slope and spread; sugar maple and red maple are a little different, with lower slopes than the other five. The four coniferous scatterplots are most similar to the two maple scatterplots, flatter and lower.

3. Soil water content is expected to vary from day to day with the most recent weather and from plot to plot depending on the topography, for example, plot elevation, exposure, drainage capacity of the sub-soil, and so on. For this question, examine the effect of the treatment assignment (`warming` vs. `ambient`) on the soil water content (`soil_water_VWC`), taking appropriate account of such variations.

- (a) First, run an analysis assuming that the treatment effect is constant over sites and over years. You should fit a suitable linear Gaussian model with soil water content – or some suitable tranformation thereof – as response. Explain how you fitted the model, show the parameter estimates with standard errors, and discuss conclusions about the effect of the treatment.

We eliminate missing water level measurements and observe that the same water level is recorded in the same plot in the same day several times for different species. We only take one such measurement for this analysis yielding 501 observations. We fit a random effects model with the assumption that the plot effect and day effect are gaussian and independent.

```
mod1=lmer(soil_water_VWC ~ warming_treatment+as.factor(year)+site+
          (1|plot_id)+(1|doy_),data=data[wuniq==1,],REML=FALSE)
ss1=summary(mod1)

print(c('warming treatment',format(ss1$coefficients[2,1:2],digits=2,nsmall=2)))

##                                Estimate          Std. Error
## "warming treatment"            "-0.031"              " 0.006"

print(c('site',format(ss1$coefficients[5,1:2],digits=2,nsmall=2)))

##              Estimate Std. Error
##      "site"      "-0.066"    " 0.011"
```

Warming treatment and site have a significant effect, with the warming treatment inducing a reduction of  $-0.03 \pm 0.006$  in water level.

The iid assumption is not compelling, especially for the temporal effects. For plot effects the inclusion of a block factor introduces correlations among plots in the same block but doesn't change the conclusions.

- (b) You should have found a small negative treatment effect in the first part of this question. Next, examine the question of whether the treatment effect is indeed constant across sites, as assumed in the first part of the question. Similarly, examine whether it is constant across years. Explain how you might address these questions, and report your answers.

We run two models with an interaction term between treatment and year, and an interaction term treatment and site. Both seem significant implying there is variation in treatment effect accross years and sites.

```
## [1] "year 2011" "0.0123"    "0.0027"
## [1] "site"      "0.039"    "0.011"
```

4. For this question we will use the data from only the `queru` species (northern red oak). The photosynthesis level of a tree may depend on both the soil water content and the treatment (in addition to variation from day-to-day weather and effects of other factors).

- (a) Build a linear Gaussian model with photosynthesis (`Asat`) – or some suitable tranformation thereof – as the response. Your analysis should accommodate block, plot, and/or temporal effects as

needed. Explain how you fitted the model, and show the parameter estimates with standard errors. Give a brief summary of the conclusions reached on the basis of the models fitted, and report a conclusion testing the following hypothesis with a likelihood ratio test: the effect (slope) of soil water content on photosynthesis level is the same for the two treatment groups.

We use the likelihood ratio test to inspect a sequence of simpler models, always using random effects for plot and day. The only significant interaction term appears to be between treatment and water content.

```
## Data: data[data$species == "queru", ]
## Models:
## mod4d: Asat ~ soil_water_VWC + warming_treatment + as.factor(year) +
## mod4d:      (1 | plot_id) + (1 | doy_)
## mod4c: Asat ~ soil_water_VWC * warming_treatment + as.factor(year) +
## mod4c:      (1 | plot_id) + (1 | doy_)
## mod4b: Asat ~ (soil_water_VWC + warming_treatment + site + as.factor(year))^2 +
## mod4b:      (1 | plot_id) + (1 | doy_)
## mod4a: Asat ~ (soil_water_VWC + warming_treatment + site + as.factor(year))^3 +
## mod4a:      (1 | plot_id) + (1 | doy_)
## mod4: Asat ~ soil_water_VWC * warming_treatment * site * as.factor(year) +
## mod4:      (1 | plot_id) + (1 | doy_)
##      Df  AIC  BIC logLik deviance Chisq Chi Df Pr(>Chisq)
## mod4d  8 1320 1348  -652    1304
## mod4c  9 1306 1337  -644    1288 16.16    1 5.8e-05 ***
## mod4b 18 1314 1377  -639    1278  9.89    9 0.36
## mod4a 25 1324 1411  -637    1274  4.19    7 0.76
## mod4  27 1327 1421  -637    1273  0.32    2 0.85
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The fixed effect coefficients are:

|                                           | Estimate | Std. Error | t value |
|-------------------------------------------|----------|------------|---------|
| ## (Intercept)                            | " 8.26"  | " 1.99"    | " 4.16" |
| ## soil_water_VWC                         | " 4.80"  | " 9.91"    | " 0.48" |
| ## warming_treatmentwarmed                | "-4.02"  | " 1.61"    | "-2.50" |
| ## as.factor(year)2010                    | " 3.31"  | " 1.45"    | " 2.28" |
| ## as.factor(year)2011                    | " 0.40"  | " 1.34"    | " 0.30" |
| ## soil_water_VWC:warming_treatmentwarmed | "40.36"  | " 9.75"    | " 4.14" |

- (b) Imagine that in the year 2010 you measured **Asat** on a red oak leaf in a new plot at site **hwrc** that was assigned to the warming treatment. Under the model you estimated what would be the mean and SD of that measurement? Provide an SE for these quantities. Show code for your calculations, and explain how you arrived at your conclusions.

We first compute the mean and variance of the water content based on the estimated model, taking into account the different variance components.

```
coef1 = ss1$coef
coef2 = ss4c$coef
soil_water_VWC_new_mean = coef1[1] + coef1[2] + coef1[3]+coef1[5]
soil_water_VWC_new_var = ss1$varcor$doy_[1]+ss1$varcor$plot_id[1]+ss1$sigma^2
```

Then we compute the prediction using the new water content mean and the variance plugging in the water content variance:

```

pred=coef2[1] + soil_water_VWC_new_mean*(coef2[2]+coef2[6]) + coef2[3] + coef2[4]
pred_SD=sqrt(soil_water_VWC_new_var*(coef2[2]+coef2[6])^2 +
             ss2$varcor$doy_[1]+ss2$varcor$plot_id[1]+ss2$sigma^2)
print(c(pred,pred_SD))

## [1] 13.2 1.6

```

Finally we can again run bootstrap to obtain SE's for these estimates.

```

##          pred pred_SD          pred pred_SD
## "PLUG" "13.78" " 4.39" "SEs" "1.50" "0.24"
##          pred pred_SD          pred pred_SD
## "NO PLUG" "13.04" " 4.46" "SEs" "1.39" "0.21"

```

- Summarize the assumptions made in the models you fitted in questions 3 and 4, and discuss the validity and implications of these assumptions.

The most problematic assumption is the independence between observations on different days. Some days are consecutive and one would expect a strong correlation between measurements on such days - as can be seen from the analysis in the next problem.

- Time permitting: Let  $Y$  be a Gaussian random vector with mean  $X\beta$ , covariance matrix  $\Sigma = W^{-1}$ , and let  $R = QY$  be the residual projection in which  $Q = I - X(X'WX)^{-1}X'W$ . Compute the mean and variance of the residual quadratic form  $R'\Delta R$  in which  $\Delta$  is a given symmetric matrix. Your answers should be expressed in terms of the matrices  $Q, \Sigma, \Delta$ .

Let  $\Delta$  be the indicator function for consecutive calendar days, and let  $Y$  be the soil water content. Compare the observed residual quadratic form with its mean and variance as computed for the fitted model in part 3(b) with two interactions. What does this calculation suggest?

The marginal covariance  $\Sigma$  is obtained from the output of the model using the random effect design matrix the diagonal matrix of variances of the random effects and the residual variance:

```

datu=data[wuniq==1,]
mod3a = lmer(soil_water_VWC ~ warming_treatment*(site+as.factor(year))+
             (1|plot_id)+(1|doy_),data=datu,REML=FALSE)
ss3a=summary(mod3a)
Z=getME(mod3a,"Z")
D=diag(c(rep(ss3a$varcor$doy_[1],49),c(rep(ss3a$varcor$plot_id[1],24))))
Sig=diag(rep(ss3a$sigma^2,501))+ Z%*%D%*%t(Z)
W=solve(Sig)
X=getME(mod3a,"X")
Wa=solve((t(X)%*%W%*%X))
Q=diag(501)-X%*%Wa%*%t(X)%*%W
R=Q%*%datu$soil_water_VWC

datf=datu$doy + (datu$year - 2009) * 365
Delta <- abs(outer(datf, datf, "-"))==1
#for (j in jj){Delta[j,j+1]=1; Delta[j+1,j]=1}

ST=unlist((t(R)%*%Delta%*%R))
A=(Delta%*%Q%*%Sig)
MEAN=unlist(sum(diag(A)))
VAR=unlist(2*sum(diag(A%*%A)))
SD=sqrt(VAR)
print(c(ST[1],MEAN,VAR,(ST[1]-MEAN)/SD))

```

```
## [1] 4.08 -0.33 0.88 4.70
```

The statistic and the related mean and standard deviation are given by:

$$S = R' \Delta R = 4.08; \quad \mu = \text{tr}(\Delta Q \Sigma) = -0.33; \quad \sigma^2 = 2\text{tr}((\Delta Q \Sigma)^2) = .88 \quad (S - \mu)/\sigma = 4.7$$